

Girishkumar Patchikoru

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PROFESSIONAL SUMMARY:

- Results-driven MLOps Engineer with 6+ years of experience building, deploying, automating, and managing scalable Machine Learning, Deep Learning, and Generative AI solutions across cloud and on-premises environments.
- Extensive experience in designing end-to-end MLOps platforms that streamline data ingestion, model training, validation, deployment, monitoring, and continuous retraining processes.
- Strong expertise in implementing CI/CD/CT pipelines for machine learning workloads using GitHub Actions, Jenkins, GitLab CI/CD, Azure DevOps, and cloud-native automation services.
- Hands-on experience with MLflow, Kubeflow, Airflow, SageMaker, Vertex AI, and Azure Machine Learning for model lifecycle management and experiment tracking.
- Proficient in deploying and managing machine learning models and Large Language Models (LLMs) on Kubernetes, Docker, OpenShift, and serverless cloud platforms.
- Experienced in building enterprise-scale Generative AI applications using LangChain, Lang Graph, Llama Index, OpenAI, Gemini, and Hugging Face ecosystems.
- Strong knowledge of Retrieval-Augmented Generation (RAG), vector databases, embeddings, semantic search, prompt engineering, and AI agent architectures.
- Extensive experience with cloud platforms including AWS, Azure, and GCP, leveraging managed AI/ML services and cloud-native architectures.
- Proficient in Infrastructure as Code (IaC) using Terraform, CloudFormation, Helm, and Ansible for repeatable and automated infrastructure provisioning.
- Strong background in distributed data processing using Apache Spark, PySpark, Kafka, Databricks, Delta Lake, and modern data lakehouse architectures.
- Expertise in building scalable model-serving architectures using REST APIs, FastAPI, Flask, Kubernetes Ingress, and service mesh technologies.
- Adept at implementing observability solutions using Prometheus, Grafana, ELK Stack, OpenTelemetry, Datadog, and cloud monitoring tools for AI platforms.

TECHNICAL SKILLS

Cloud Platforms: AWS (SageMaker, Bedrock, EC2, EKS, Lambda, S3, IAM, VPC, CloudWatch), Azure (Azure ML, AKS, ADF, Azure DevOps), GCP (Vertex AI, GKE, Big Query, Cloud Storage)

MLOps Platforms: MLflow, Kubeflow, Amazon SageMaker, Vertex AI, Azure Machine Learning, Flyte

Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Hugging Face Transformers

Generative AI & LLMops: OpenAI API, LangChain, Lang Graph, Llama Index, RAG, Prompt Engineering, Vector Databases, AI Agents, Fine-Tuning, Model Evaluation

Vector Databases: Pinecone, Weaviate, Chroma DB, FAISS

Data Engineering: Apache Spark, Kafka, Airflow, Delta Lake, Databricks, Feature Store,

Containerization & Orchestration: Docker, Kubernetes (EKS/GKE/AKS), Helm, ArgoCD, Istio

CI/CD & Automation: Jenkins, GitHub Actions, GitLab CI/CD, Azure DevOps, Argo Workflows

Infrastructure as Code: Terraform, CloudFormation, Ansible

Monitoring & Observability: Prometheus, Grafana, ELK Stack, CloudWatch, Evidently AI, Open Telemetry

Programming: Python, SQL, Bash, Shell Scripting

Security & DevSecOps: HashiCorp Vault, IAM, RBAC, Snyk, Trivy, SonarQube, Secrets Management

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis

Operating Systems: Linux (RHEL, Ubuntu, CentOS)

PROFESSIONAL EXPERIENCE:

Client: MUFG, New York, NY

July 2024 – Till Date

Role: MLOps Engineer

Responsibilities

- Designed, developed, and maintained end-to-end Machine Learning Operations (MLOps) pipelines for data ingestion, feature engineering, model training, validation, deployment, and monitoring.

- Automated ML workflows using Python, MLflow, Kubeflow, Apache Airflow, and cloud-native orchestration tools to improve scalability and operational efficiency.
- Built and managed CI/CD/CT pipelines for machine learning models using Jenkins, GitHub Actions, GitLab CI/CD, and Azure DevOps.
- Containerized ML applications using Docker and deployed scalable inference services on Kubernetes, OpenShift, and cloud-managed Kubernetes platforms.
- Implemented model versioning, experiment tracking, artifact management, and model registry solutions using MLflow and enterprise MLOps platforms.
- Collaborated closely with Data Scientists, Data Engineers, and Software Development teams to operationalize machine learning models in production environments.
- Developed and optimized batch and real-time ML data pipelines using Apache Spark, PySpark, Kafka, Databricks, and cloud data services.
- Managed cloud-based AI/ML infrastructure across AWS, Azure, and GCP, leveraging services such as SageMaker, Vertex AI, Azure Machine Learning, EKS, AKS, and GKE.
- Implemented Infrastructure as Code (IaC) using Terraform, CloudFormation, and Ansible to automate environment provisioning and configuration management.
- Established automated model deployment strategies including blue-green deployments, canary releases, and rollback mechanisms.
- Built feature engineering pipelines and integrated feature stores to improve model consistency, reproducibility, and governance.
- Implemented model monitoring solutions to track prediction quality, model drift, data drift, latency, and system health in production.
- Developed observability frameworks using Prometheus, Grafana, ELK Stack, OpenTelemetry, Datadog, and Splunk.
- Designed secure AI/ML platforms using IAM, RBAC, OAuth2, JWT, HashiCorp Vault, encryption, and secrets management best practices.

Environment: AWS SageMaker, Bedrock, MLflow, Kubeflow, Airflow, Python, Spark, Docker, Kubernetes (EKS), Jenkins, GitHub Actions, GitLab CI/CD, ArgoCD, Terraform, Prometheus, Grafana, OpenAI, LangChain, Pinecone, Chroma DB.

Client: Icertis Solution Pvt. Ltd

Feb 2022 – Aug 2023

Role: MLOps Engineer

Responsibilities:

- Designed and implemented end-to-end MLOps pipelines supporting model training, validation, deployment, monitoring, and automated retraining workflows.
- Built scalable machine learning platforms using Amazon SageMaker, MLflow, Kubeflow, and Kubernetes to support enterprise AI initiatives.
- Automated ML model deployment pipelines using Jenkins, GitHub Actions, GitLab CI/CD, ArgoCD, and Infrastructure as Code practices.
- Developed Continuous Training (CT) and Continuous Deployment (CD) frameworks for machine learning models, reducing deployment time and improving model reliability.
- Deployed and managed Large Language Models (LLMs) and Generative AI applications using OpenAI, Bedrock, Hugging Face, and LangChain frameworks.
- Implemented Retrieval-Augmented Generation (RAG) solutions integrating vector databases such as Pinecone, ChromaDB, and FAISS for enterprise knowledge search systems.
- Built feature engineering and data processing pipelines using Apache Spark, Airflow, Kafka, and AWS services to support real-time inference workloads.
- Managed Kubernetes-based AI workloads, including GPU-enabled clusters, autoscaling, model serving, and distributed training environments.
- Integrated MLflow Model Registry and experiment tracking to improve model versioning, governance, and reproducibility.
- Implemented model monitoring solutions for data drift, concept drift, model performance degradation, and inference latency tracking.

- Developed AI observability dashboards using Prometheus, Grafana, CloudWatch, and ELK Stack for proactive monitoring of ML systems.
- Automated infrastructure provisioning using Terraform and Ansible to support scalable and secure MLOps environments.
- Implemented enterprise-grade security controls using HashiCorp Vault, IAM, RBAC, secrets rotation, and compliance automation.
- Collaborated with Data Scientists and ML Engineers to operationalize machine learning models and accelerate AI product delivery.

Environment: Amazon SageMaker, MLflow, Kubeflow, and Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, ArgoCD, OpenAI, Bedrock, Hugging Face, and LangChain frameworks, Apache Spark, Airflow, Kafka, and AWS.

Client: Adrian Technologies Pvt Ltd

Apr 2020 – Jan 2022

Role: DevOps Engineer

Responsibilities:

- Developed and maintained scalable CI/CD pipelines using **Jenkins and GitLab CI**, automating build, test, and deployment processes for enterprise SaaS applications, ensuring faster and more reliable software delivery
- Containerized applications using **Docker** and orchestrated deployments on **Kubernetes clusters**, enabling horizontal scalability and improved application resilience
- Provisioned and managed cloud infrastructure on AWS using **Terraform**, creating reusable modules for consistent and repeatable infrastructure deployments across multiple environments
- Implemented comprehensive monitoring and alerting systems using **Prometheus and Grafana**, enabling proactive issue detection and performance optimization
- Automated configuration management using **Ansible**, standardizing environment setup and reducing configuration drift across development and production systems
- Designed and implemented multi-environment deployment strategies (Dev, QA, UAT, Prod) with controlled promotion pipelines and approval workflows
- Integrated automated testing and rollback mechanisms into CI/CD pipelines to improve deployment success rates and minimize downtime during failures
- Managed version control workflows using **Git**, implementing branching strategies and code review practices to ensure code quality and collaboration
- Strengthened cloud security by implementing **IAM policies, role-based access control, and secrets management solutions**, ensuring compliance with enterprise standards
- Collaborated closely with development, QA, and product teams in Agile environments to streamline release cycles and improve delivery efficiency

Environment: AWS, Terraform, Docker, Kubernetes, Jenkins, GitLab CI, Ansible, Prometheus, Grafana, Git, Linux, Bash

EDUCATION

Texas A&M University–Corpus Christi

Masters in computer Science

Certifications:

- [Google Professional Cloud Architect Certification](#)
- [Microsoft Certified: Azure Developer Associate](#)
- [Oracle Cloud Infrastructure 2025 Certified Foundations Associate](#)
- [Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate](#)
- [Prompt Engineering & Programming with OpenAI](#)
- [Google Data Analytics Professional Certificate](#)